

SECTION 2 PROCESSING STATES

This section describes the processing states of the MC68020/EC020. It describes the functions of the bits in the supervisor portion of the SR and the actions taken by the processor in response to exception conditions.

Unless the processor has halted, it is always in either the normal or the exception processing state. Whenever the processor is executing instructions or fetching instructions or operands, it is in the normal processing state. The processor is also in the normal processing state while it is storing instruction results or communicating with a coprocessor.

NOTE

Exception processing refers specifically to the transition from normal processing of a program to normal processing of system routines, interrupt routines, and other exception handlers. Exception processing includes all stacking operations, the fetch of the exception vector, and the filling of the instruction pipe caused by an exception. Exception processing has completed when execution of the first instruction of the exception handler routine begins.

The processor enters the exception processing state when an interrupt is acknowledged, when an instruction is traced or results in a trap, or when some other exception condition arises. Execution of certain instructions or unusual conditions occurring during the execution of any instruction can cause exceptions. External conditions, such as interrupts, bus errors, and some coprocessor responses, also cause exceptions. Exception processing provides an efficient transfer of control to handlers and routines that process the exceptions.

A catastrophic system failure occurs whenever the processor receives a bus error or generates an address error while in the exception processing state. This type of failure halts the processor. For example, if during the exception processing of one bus error another bus error occurs, the MC68020/EC020 has not completed the transition to normal processing and has not completed saving the internal state of the machine; therefore, the processor assumes that the system is not operational and halts. Only an external reset can restart a halted processor. (When the processor executes a STOP instruction, it is in a special type of normal processing state—one without bus cycles. It is stopped, not halted.)

2.1 PRIVILEGE LEVELS

The processor operates at one of two privilege levels: the user level or the supervisor level. The supervisor level has higher privileges than the user level. Not all processor or coprocessor instructions are permitted to execute at the lower privileged user level, but all are available at the supervisor level. This arrangement allows a separation of supervisor and user so the supervisor can protect system resources from uncontrolled access. The S-bit in the SR is used to select either the user or supervisor privilege level and either the USP or an SSP for stack operations. The processor identifies a bus access (supervisor or user mode) via the function codes so that differentiation between supervisor level and user level can be maintained.

In many systems, the majority of programs execute at the user level. User programs can access only their own code and data areas and can be restricted from accessing other information. The operating system typically executes at the supervisor privilege level. It has access to all resources, performs the overhead tasks for the user-level programs, and coordinates user-level program activities.

2.1.1 Supervisor Privilege Level

The supervisor level is the higher privilege level. The privilege level is determined by the S-bit of the SR; if the S-bit is set, the supervisor privilege level applies, and all instructions are executable. The bus cycles for instructions executed at the supervisor level are normally classified as supervisor references, and the values of the FC2–FC0 signals refer to supervisor address spaces.

In a multitasking operating system, it is more efficient to have a supervisor stack space associated with each user task and a separate stack space for interrupt-associated tasks. The MC68020/EC020 provides two supervisor stacks, master and interrupt; the M bit of the SR selects which of the two is active. When the M-bit is set, references to the SSP implicitly or to address register seven (A7) explicitly, access the MSP. The operating system sets the MSP for each task to point to a task-related area of supervisor data space. This arrangement separates task-related supervisor activity from asynchronous, I/O-related supervisor tasks that may be only coincidental to the currently executing task. The MSP can separately maintain task control information for each currently executing user task, and the software updates the MSP when a task switch is performed, providing an efficient means for transferring task-related stack items. The other supervisor stack pointer, the ISP, can be used for interrupt control information and workspace area as interrupt handling routines require.

When the M-bit is clear, the MC68020/EC020 is in the interrupt mode of the supervisor privilege level, and operation is the same as supervisor mode in the MC68000, MC68HC001, MC68008, and MC68010. (The processor is in this mode after a reset operation.) All SSP references access the ISP in this mode.

The value of the M-bit in the SR does not affect execution of privileged instructions; both master and interrupt modes are at the supervisor privilege level. Instructions that affect the M-bit are MOVE to SR, ANDI to SR, EORI to SR, ORI to SR, and RTE. Also, the processor automatically saves the M-bit value and clears it in the SR as part of exception processing for interrupts.

All exception processing is performed at the supervisor privilege level. All bus cycles generated during exception processing are supervisor references, and all stack accesses use the active SSP.

2.1.2 User Privilege Level

The user level is the lower privilege level. The privilege level is determined by the S-bit of the SR; if the S-bit is clear, the processor executes instructions at the user privilege level.

Most instructions execute at either privilege level, but some instructions that have important system effects are privileged and can only be executed at the supervisor level. For instance, user programs are not allowed to execute the STOP instruction or the RESET instruction. To prevent a user program from entering the supervisor privilege level except in a controlled manner, instructions that can alter the S-bit in the SR are privileged. The TRAP #n instruction provides controlled access to operating system services for user programs.

The bus cycles for an instruction executed at the user privilege level are classified as user references, and the values of the FC2–FC0 signals specify user address spaces. While the processor is at the user level, references to the system stack pointer implicitly, or to address register seven (A7) explicitly, refer to the USP.

2.1.3 Changing Privilege Level

To change from the user to the supervisor privilege level, one of the conditions that causes the processor to perform exception processing must occur. This causes a change from the user level to the supervisor level and can cause a change from the master mode to the interrupt mode. Exception processing saves the current values of the S and M bits of the SR (along with the rest of the SR) on the active supervisor stack, and then sets the S-bit, forcing the processor into the supervisor privilege level. When the exception being processed is an interrupt and the M-bit is set, the M-bit is cleared, putting the processor into the interrupt mode. Execution of instructions continues at the supervisor level to process the exception condition.

To return to the user privilege level, a system routine must execute one of the following instructions: MOVE to SR, ANDI to SR, EORI to SR, ORI to SR, or RTE. These instructions execute at the supervisor privilege level and can modify the S-bit of the SR. After these instructions execute, the instruction pipeline is flushed and is refilled from the appropriate address space.

The RTE instruction returns to the program that was executing when the exception occurred. It restores the exception stack frame saved on the supervisor stack. If the frame

on top of the stack was generated by an interrupt, trap, or instruction exception, the RTE instruction restores the SR and PC to the values saved on the supervisor stack. The processor then continues execution at the restored PC address and at the privilege level determined by the S-bit of the restored SR. If the frame on top of the stack was generated by a bus fault (bus error or address error exception), the RTE instruction restores the entire saved processor state from the stack.

2.2 ADDRESS SPACE TYPES

The processor specifies a target address space for every bus cycle with the FC2–FC0 signals according to the type of access required. In addition to distinguishing between supervisor/user and program/data, the processor can identify special processor cycles, such as the interrupt acknowledge cycle, and the memory management unit can control accesses and translate addresses appropriately. Table 2-1 lists the types of accesses defined for the MC68020/EC020 and the corresponding values of the FC2–FC0 signals.

Table 2-1. Address Space Encodings

FC2	FC1	FC0	Address Space
0	0	0	(Undefined, Reserved)*
0	0	1	User Data Space
0	1	0	User Program Space
0	1	1	(Undefined, Reserved)*
1	0	0	(Undefined, Reserved)*
1	0	1	Supervisor Data Space
1	1	0	Supervisor Program Space
1	1	1	CPU Space

* Address space 3 is reserved for user definition; 0 and 4 are reserved for future use by Motorola.

The memory locations of user program and data accesses are not predefined; neither are the locations of supervisor data space. During reset, the first two long words beginning at memory location zero in the supervisor program space are used for processor initialization. No other memory locations are explicitly defined by the MC68020/EC020.

A function code of \$7 selects the CPU address space. This is a special address space that does not contain instructions or operands but is reserved for special processor functions. The processor uses accesses in this space to communicate with external devices for special purposes. For example, all M68000 processors use the CPU space for interrupt acknowledge cycles. The MC68020/EC020 also generate CPU space accesses for breakpoint acknowledge and coprocessor operations.

Supervisor programs can use the MOVES instruction to access all address spaces, including the user spaces and the CPU address space. Although the MOVES instruction can be used to generate CPU space cycles, this may interfere with proper system operation. Thus, the use of MOVES to access the CPU space should be done with caution.

2.3 EXCEPTION PROCESSING

An exception is defined as a special condition that preempts normal processing. Both internal and external conditions can cause exceptions. External conditions that cause exceptions are interrupts from external devices, bus errors, coprocessor-detected errors, and reset. Instructions, address errors, tracing, and breakpoints are internal conditions that cause exceptions. The TRAP, TRAPcc, TRAPV, cpTRAPcc, CHK, CHK2, RTE, BKPT, CALLM, RTM, cp RESTORE, DIVS and DIVU instructions can generate exceptions as part of their normal execution. In addition, illegal instructions, privilege violations, and coprocessor protocol violations cause exceptions.

Exception processing, which is the transition from the normal processing of a program to the processing required for the exception condition, involves the exception vector table and an exception stack frame. The following paragraphs describe the exception vectors and a generalized exception stack frame. Exception processing is discussed in detail in **Section 6 Exception Processing**. Coprocessor-detected exceptions are discussed in detail in **Section 7 Coprocessor Interface Description**.

2.3.1 Exception Vectors

The VBR contains the base address of the 1024-byte exception vector table, which consists of 256 exception vectors. Exception vectors contain the memory addresses of routines that begin execution at the completion of exception processing. These routines perform a series of operations appropriate for the corresponding exceptions. Because the exception vectors contain memory addresses, each consists of one long word, except for the reset vector. The reset vector consists of two long words: the address used to initialize the ISP and the address used to initialize the PC.

The address of an exception vector is derived from an 8-bit vector number and the VBR. The vector numbers for some exceptions are obtained from an external device; others are supplied automatically by the processor. The processor multiplies the vector number by four to calculate the vector offset, which it adds to the VBR. The sum is the memory address of the vector. All exception vectors are located in supervisor data space, except the reset vector, which is located in supervisor program space. Only the initial reset vector is fixed in the processor's memory map; once initialization is complete, there are no fixed assignments. Since the VBR provides the base address of the vector table, the vector table can be located anywhere in memory; it can even be dynamically relocated for each task that is executed by an operating system. Details of exception processing are provided in **Section 6 Exception Processing**, and Table 6-1 lists the exception vector assignments.

2.3.2 Exception Stack Frame

Exception processing saves the most volatile portion of the current processor context on the top of the supervisor stack. This context is organized in a format called the exception stack frame. This information always includes a copy of the SR, the PC, the vector offset of the vector, and the frame format field. The frame format field identifies the type of stack frame. The RTE instruction uses the value in the format field to properly restore the information stored in the stack frame and to deallocate the stack space. The general form of the exception stack frame is illustrated in Figure 2-1. Refer to **Section 6 Exception Processing** for a complete list of exception stack frames.

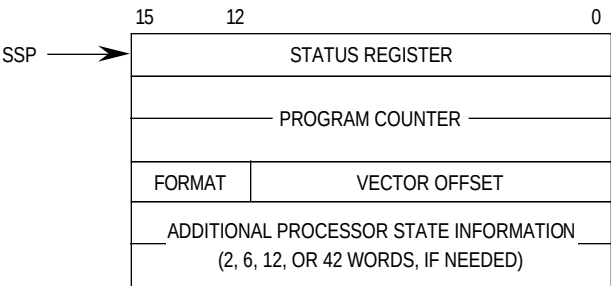


Figure 2-1. General Exception Stack Frame